

```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

```
In [3]: data=pd.read_csv('Iris.csv')
data
```

```
Out[3]:
```

|     | sepal.length | sepal.width | petal.length | petal.width | variety   |
|-----|--------------|-------------|--------------|-------------|-----------|
| 0   | 5.1          | 3.5         | 1.4          | 0.2         | Setosa    |
| 1   | 4.9          | 3.0         | 1.4          | 0.2         | Setosa    |
| 2   | 4.7          | 3.2         | 1.3          | 0.2         | Setosa    |
| 3   | 4.6          | 3.1         | 1.5          | 0.2         | Setosa    |
| 4   | 5.0          | 3.6         | 1.4          | 0.2         | Setosa    |
| ... | ...          | ...         | ...          | ...         | ...       |
| 145 | 6.7          | 3.0         | 5.2          | 2.3         | Virginica |
| 146 | 6.3          | 2.5         | 5.0          | 1.9         | Virginica |
| 147 | 6.5          | 3.0         | 5.2          | 2.0         | Virginica |
| 148 | 6.2          | 3.4         | 5.4          | 2.3         | Virginica |
| 149 | 5.9          | 3.0         | 5.1          | 1.8         | Virginica |

150 rows × 5 columns

```
In [4]: data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   sepal.length    150 non-null   float64
1   sepal.width     150 non-null   float64
2   petal.length    150 non-null   float64
3   petal.width     150 non-null   float64
4   variety         150 non-null   str
dtypes: float64(4), str(1)
memory usage: 7.2 KB
```

```
In [5]: data.describe()
```

Out[5]:

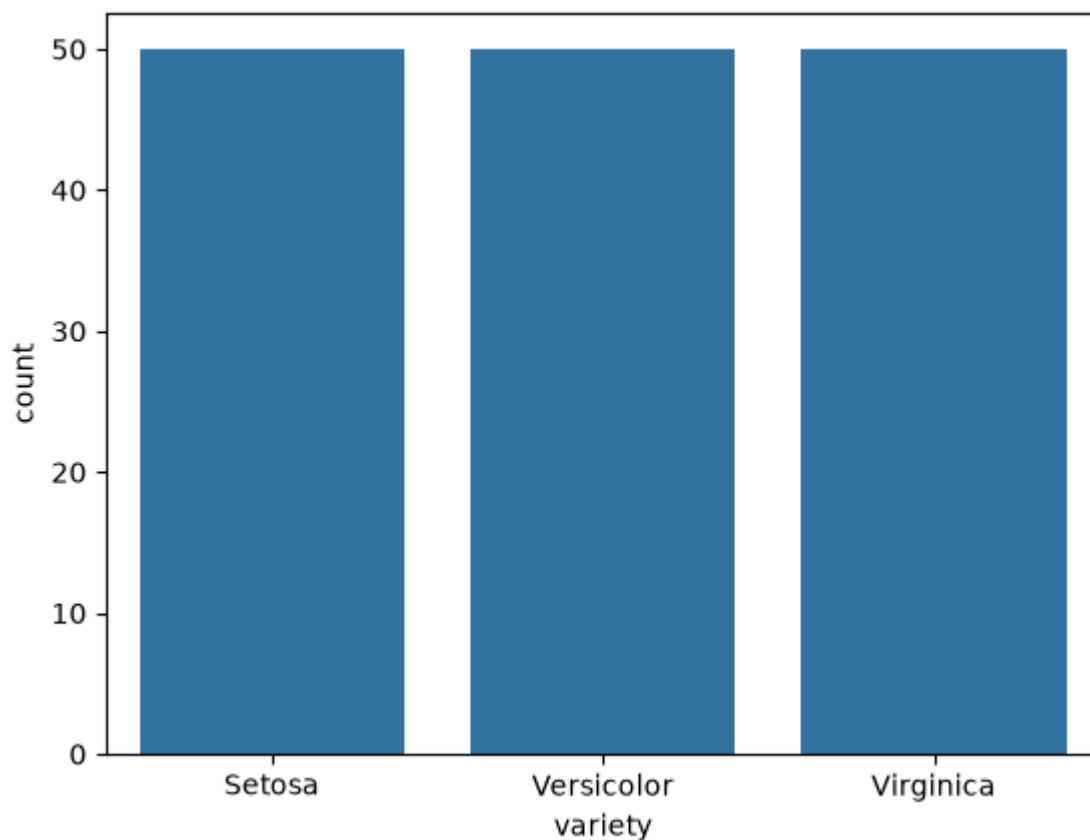
|              | sepal.length | sepal.width | petal.length | petal.width |
|--------------|--------------|-------------|--------------|-------------|
| <b>count</b> | 150.000000   | 150.000000  | 150.000000   | 150.000000  |
| <b>mean</b>  | 5.843333     | 3.057333    | 3.758000     | 1.199333    |
| <b>std</b>   | 0.828066     | 0.435866    | 1.765298     | 0.762238    |
| <b>min</b>   | 4.300000     | 2.000000    | 1.000000     | 0.100000    |
| <b>25%</b>   | 5.100000     | 2.800000    | 1.600000     | 0.300000    |
| <b>50%</b>   | 5.800000     | 3.000000    | 4.350000     | 1.300000    |
| <b>75%</b>   | 6.400000     | 3.300000    | 5.100000     | 1.800000    |
| <b>max</b>   | 7.900000     | 4.400000    | 6.900000     | 2.500000    |

In [6]: `data.value_counts('variety')`

Out[6]:

|            |    |
|------------|----|
| variety    |    |
| Setosa     | 50 |
| Versicolor | 50 |
| Virginica  | 50 |

Name: count, dtype: int64

In [7]: `sns.countplot(x='variety',data=data,)  
plt.show()`

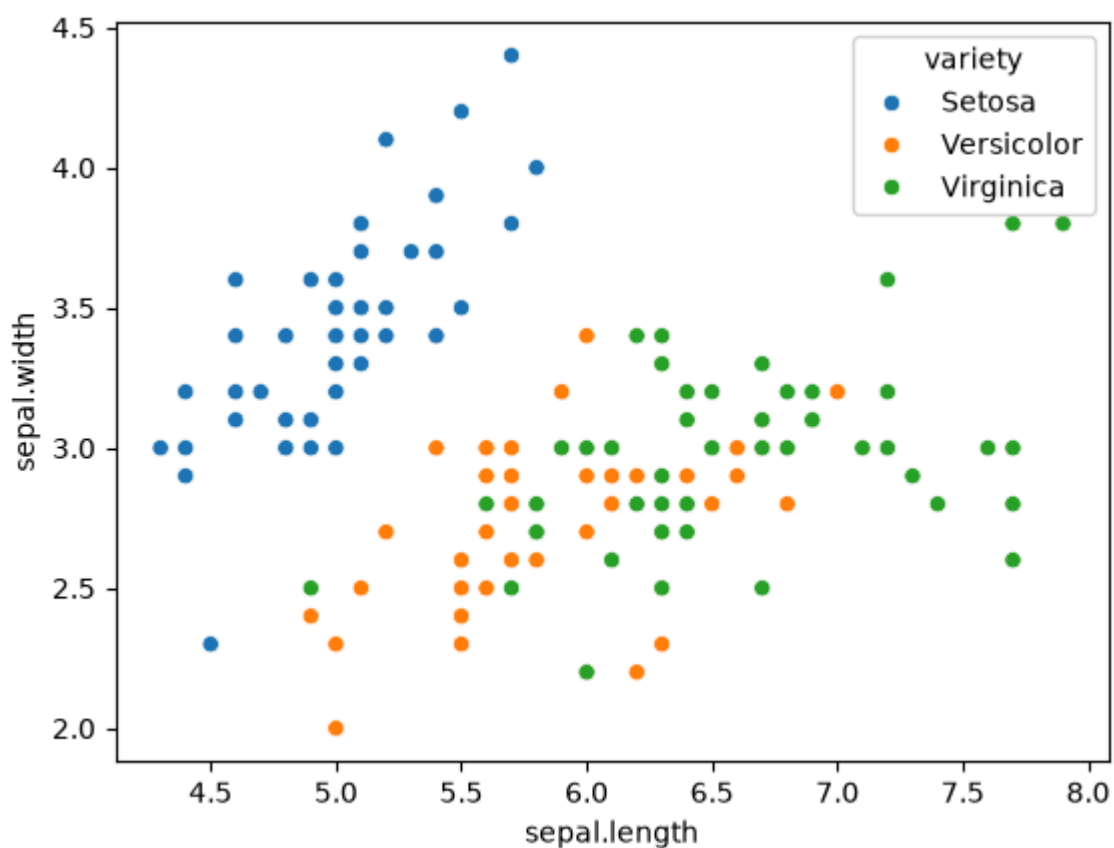
In [8]: `dummies=pd.get_dummies(data.variety)  
FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axis=1  
FinalDataset.head()`

```
Out[8]:
```

|   | Setosa | Versicolor | Virginica | sepal.length | sepal.width | petal.length | petal.width |
|---|--------|------------|-----------|--------------|-------------|--------------|-------------|
| 0 | True   | False      | False     | 5.1          | 3.5         | 1.4          | 0.2         |
| 1 | True   | False      | False     | 4.9          | 3.0         | 1.4          | 0.2         |
| 2 | True   | False      | False     | 4.7          | 3.2         | 1.3          | 0.2         |
| 3 | True   | False      | False     | 4.6          | 3.1         | 1.5          | 0.2         |
| 4 | True   | False      | False     | 5.0          | 3.6         | 1.4          | 0.2         |

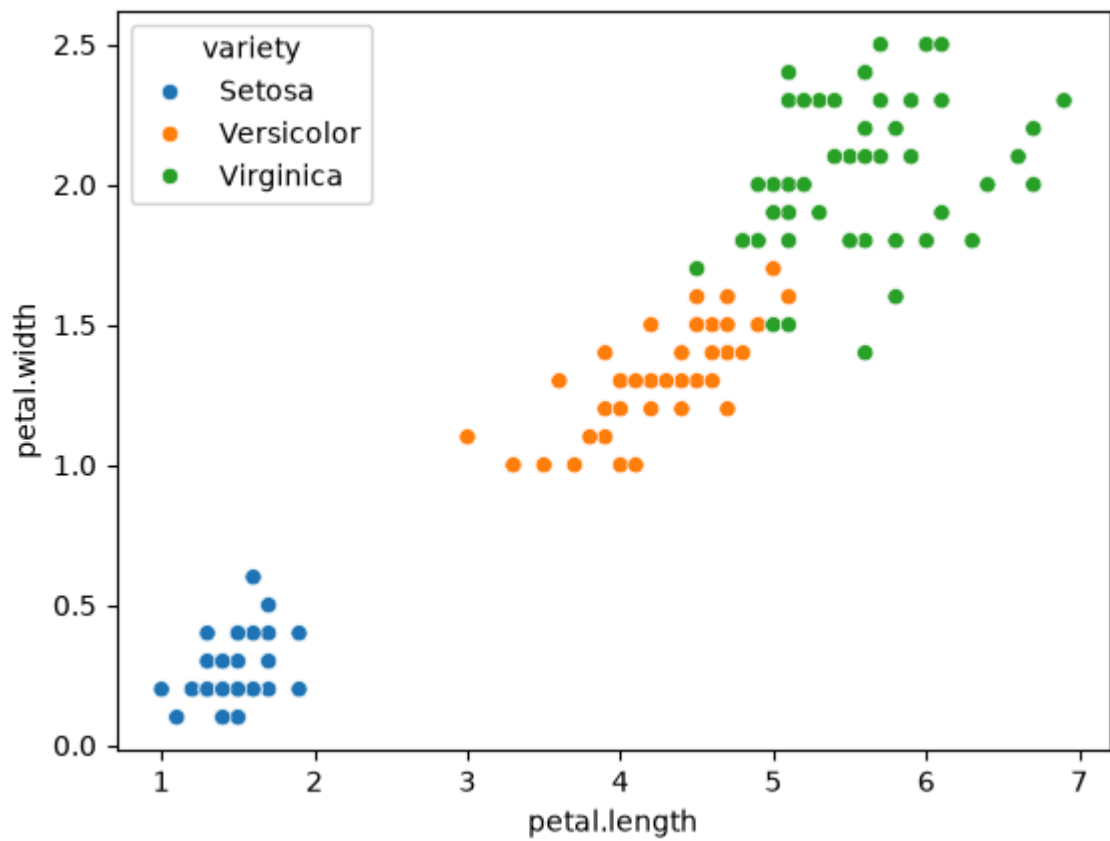
```
In [11]: sns.scatterplot(x='sepal.length',y='sepal.width',hue='variety',data=data)
```

```
Out[11]: <Axes: xlabel='sepal.length', ylabel='sepal.width'>
```

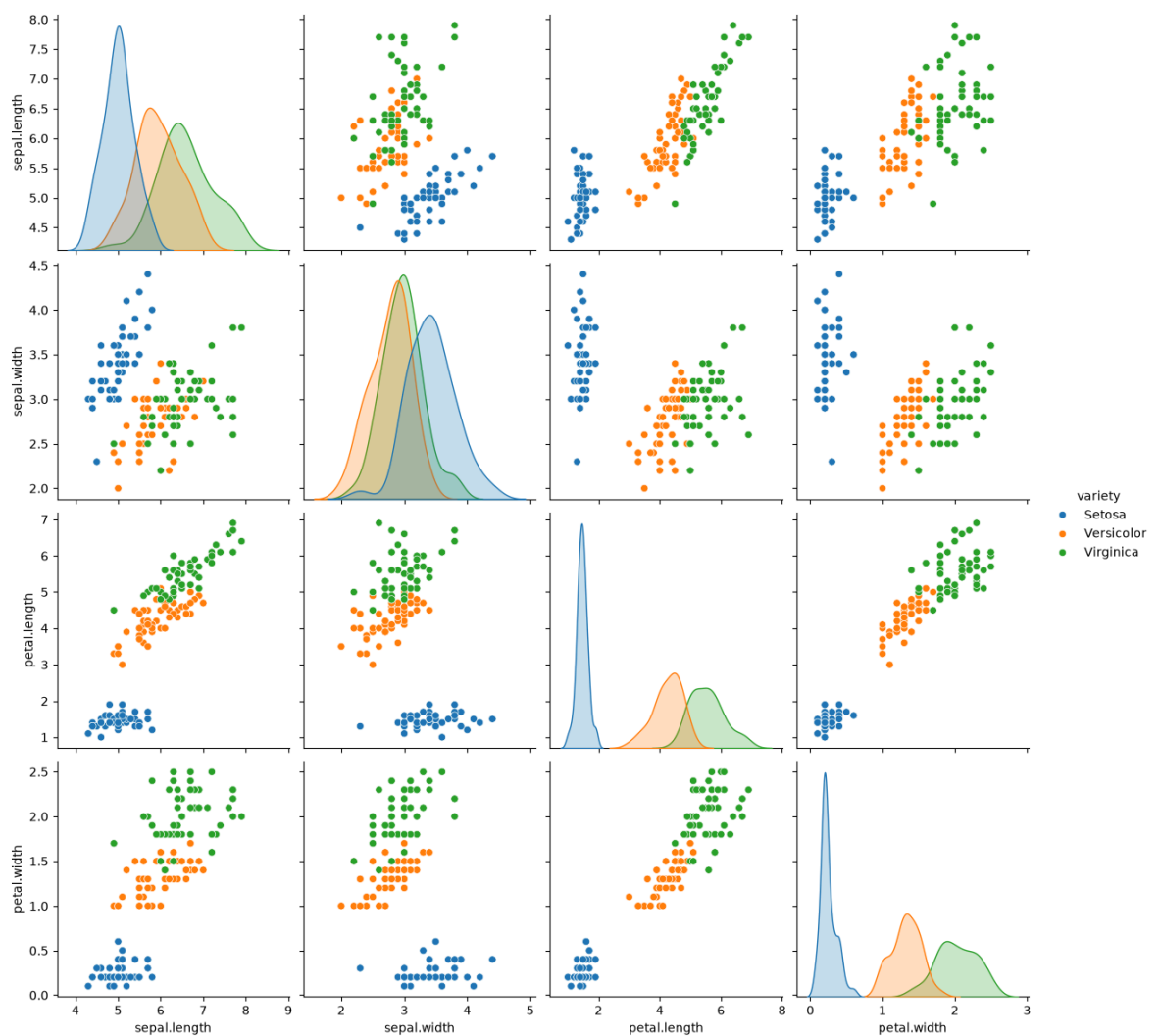


```
In [13]: sns.scatterplot(x='petal.length',y='petal.width',hue='variety',data=data)
```

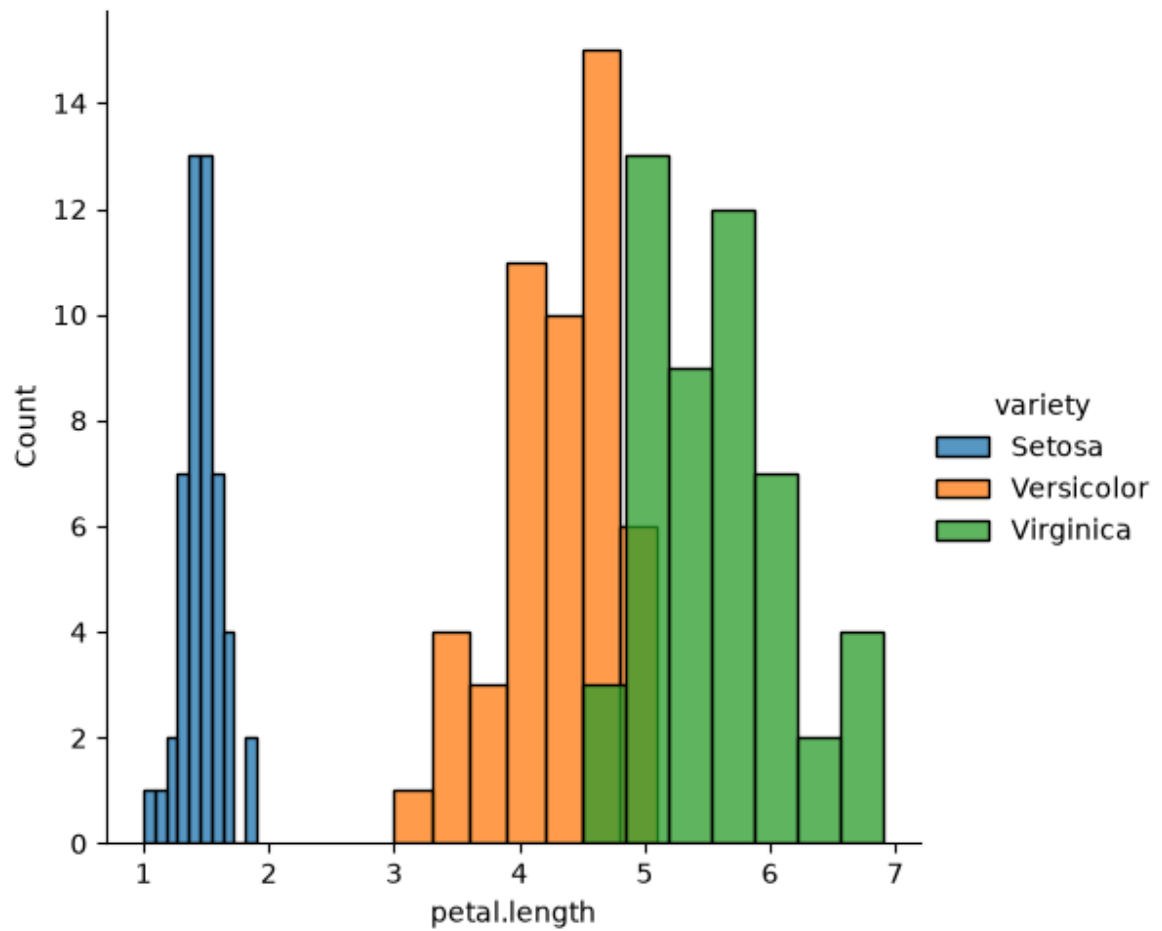
```
Out[13]: <Axes: xlabel='petal.length', ylabel='petal.width'>
```



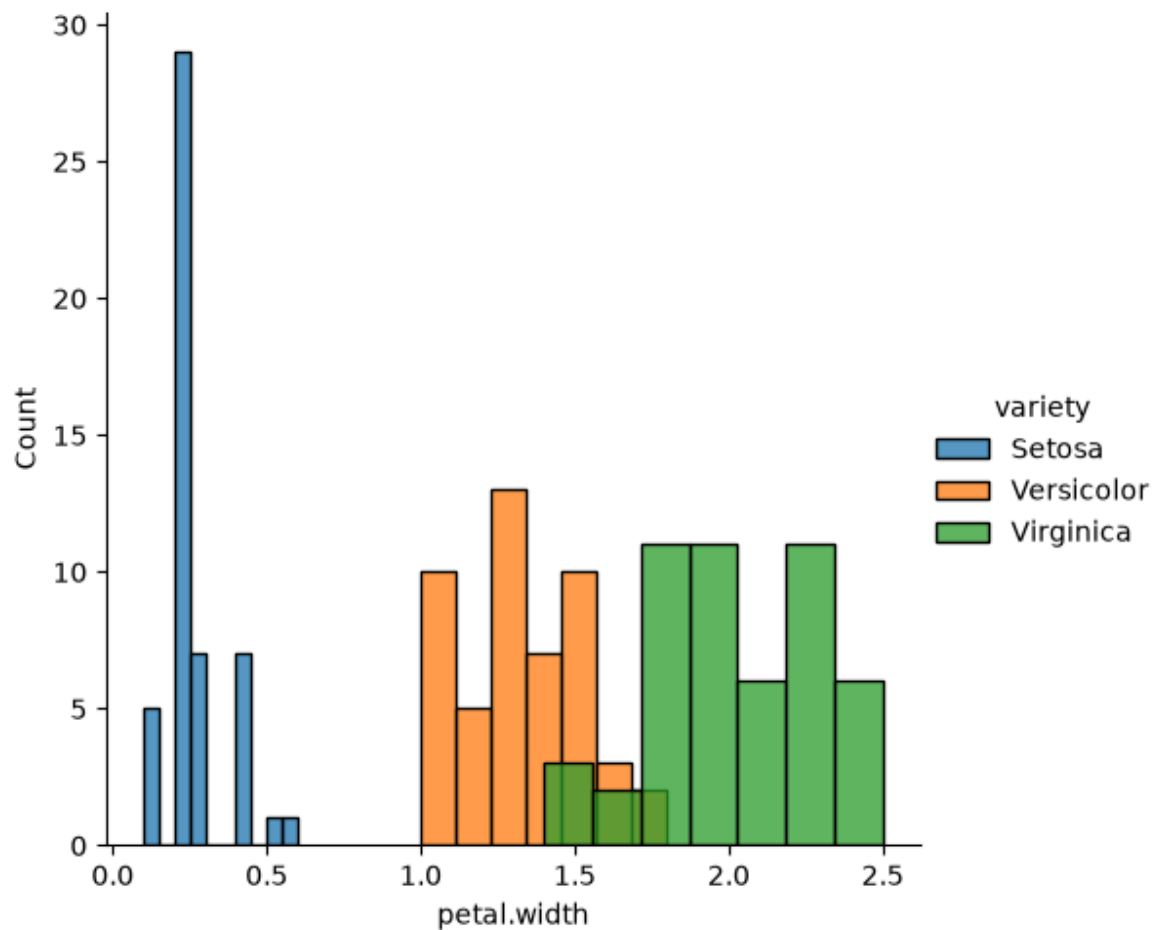
```
In [14]: sns.pairplot(data,hue='variety',height=3);
```



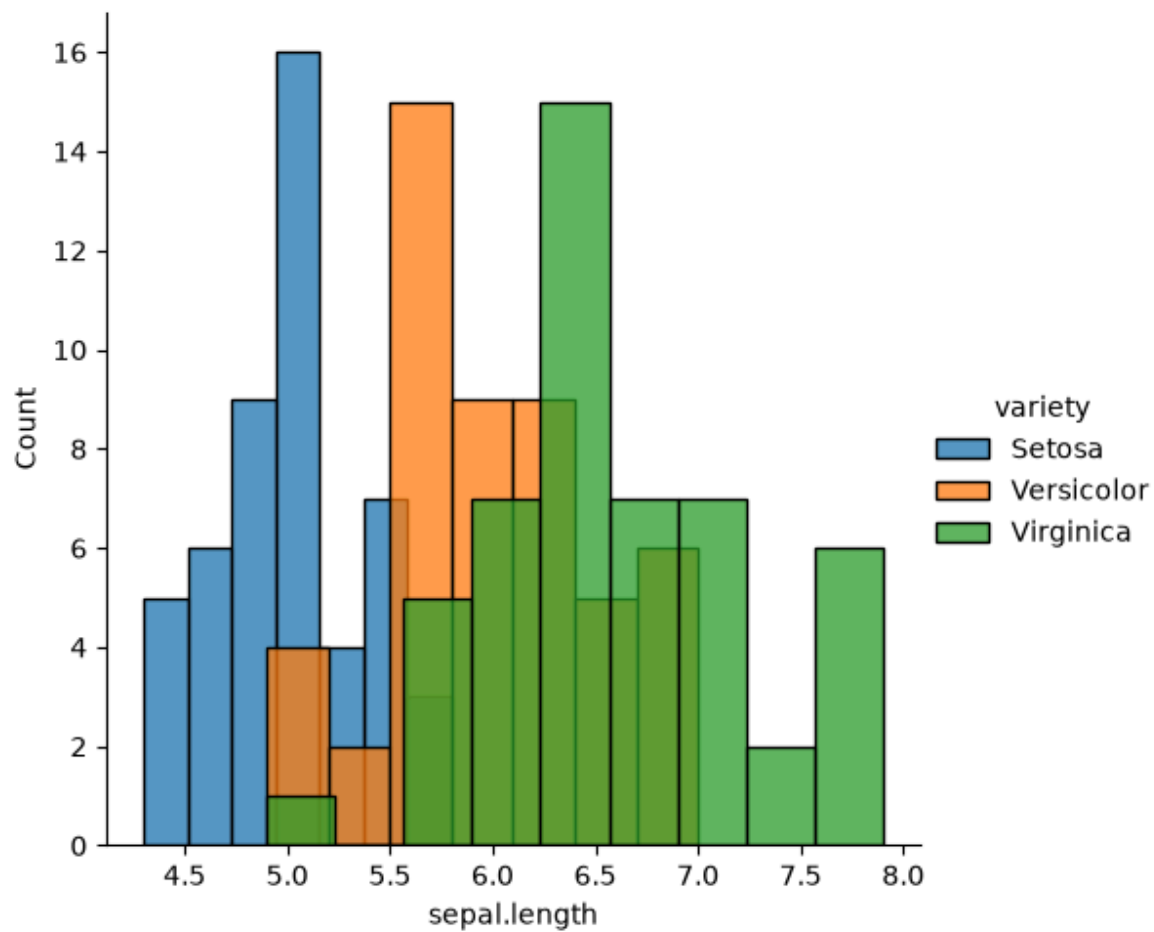
```
In [17]: sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'petal.length').add_
```



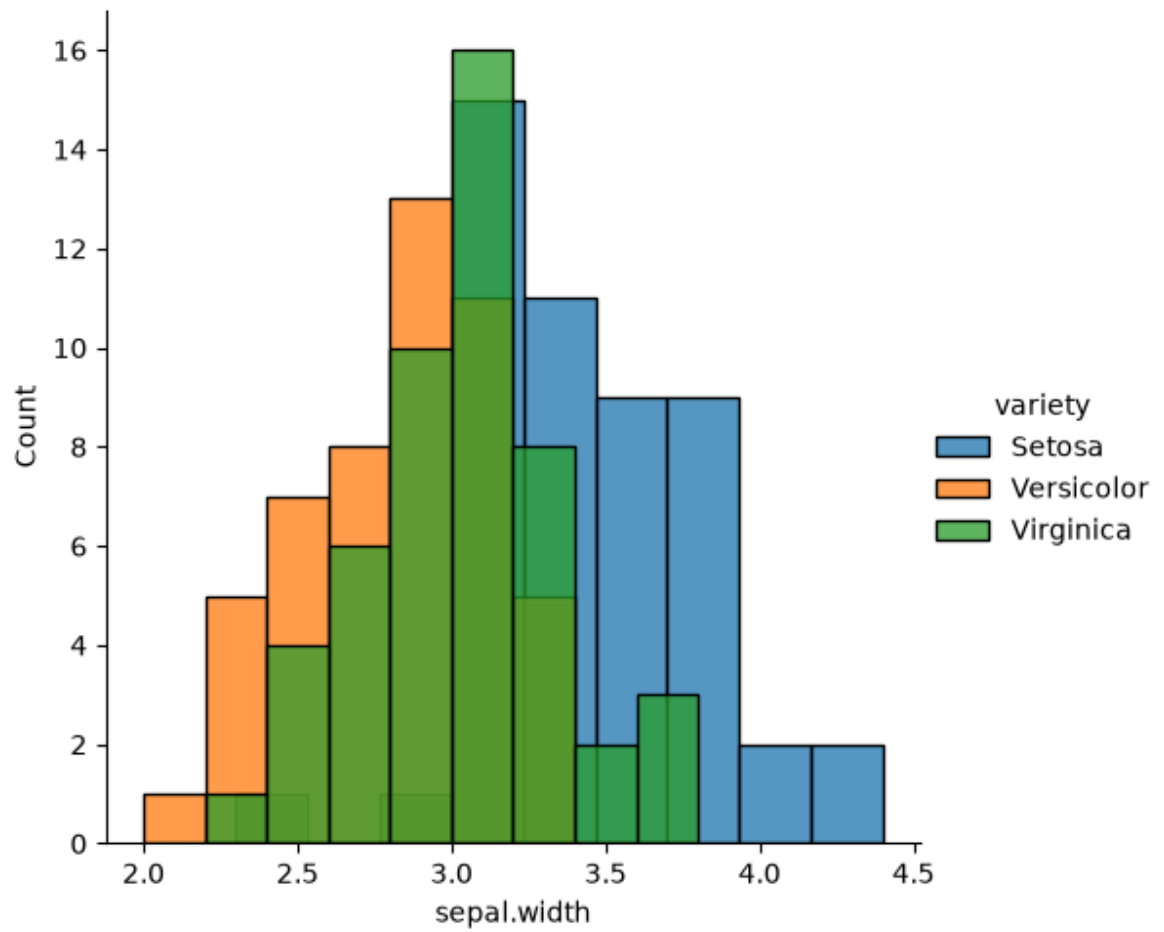
```
In [19]: sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'petal.width').add_1  
plt.show();
```



```
In [20]: sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'sepal.length').add_
plt.show();
```



```
In [21]: sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'sepal.width').add_1  
plt.show();
```



In [ ]: